How fast does opening two windows replace the air in a room?

Stoßlüften: CFD dataset for an ML surrogate model · Machine Learning, Hochschule Hannover · status 15 Oct 2026

Goal

We simulate window airing (*Stoßlüften*) on the computer with CFD (computational fluid dynamics: a simulation of how air flows) and collect the results in a table. A machine-learning model will later learn from this table, so that it gives the answer instantly instead of running a slow simulation.

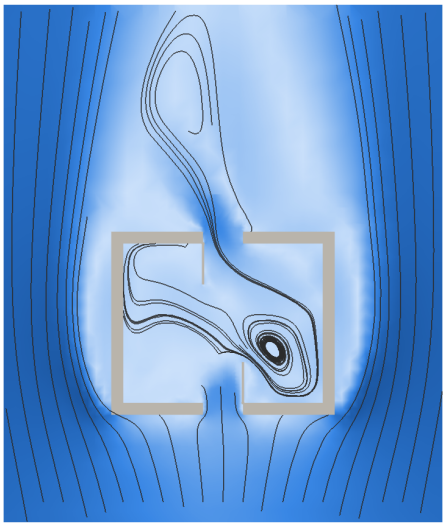
Inputs (ML features)

- window layout: same, opposite or adjacent walls
- position of each window on its wall
- turned open or tilted, and how far
- wind speed, 1–5 m/s

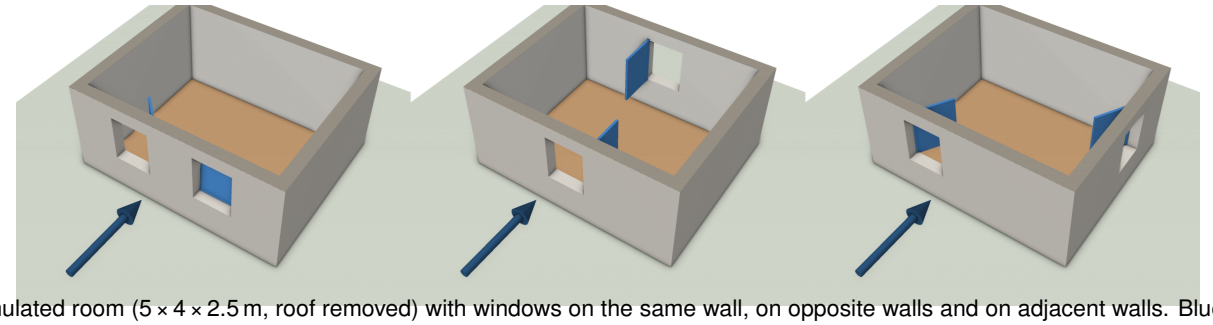
Outputs (ML targets)

- share of the room air replaced, every 30 s for 10 min
- time until 90 % of the air is replaced

Example: windows on opposite walls, fully open, wind 3 m/s: 90 % of the air is replaced after 76 s (right). With both windows on the same wall it takes 7–10 min, because the wind cannot blow straight through the room.

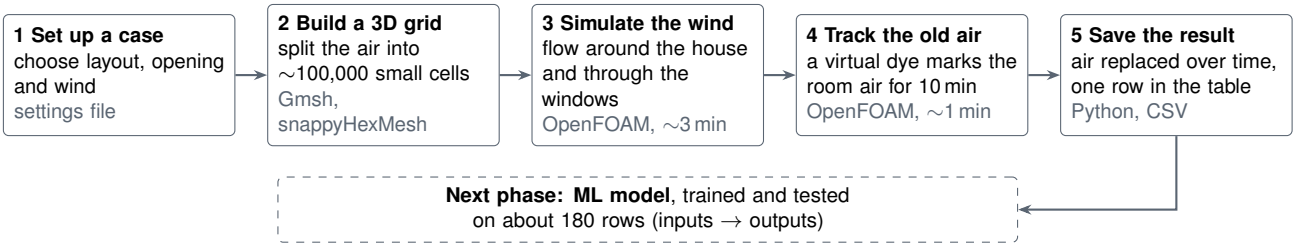


Example case seen from above at window height. The wind (3 m/s, from below) enters through one window, crosses the room as a jet and leaves through the opposite window. Colour: air speed from 0 (light) to 4 m/s (dark); lines: flow paths.



The simulated room (5 × 4 × 2.5 m, roof removed) with windows on the same wall, on opposite walls and on adjacent walls. Blue plates: opened windows. Arrow: wind.

Process: one simulation, fully automatic



Checks before the big run

A simulation can give wrong numbers that still look reasonable, so we tested how far the results can be trusted.

Grid size	Finer or coarser cells change the results by less than 2.5 % (our first grid type: 10–20 %, so we replaced it). Cases with both windows on the same wall: up to 17 %, an extra check is planned.
Small changes	Moving a window by 1 cm changes the result by less than 2.5 %.
Fast vs. exact method 12 cases	Averaged wind (∼5 min per case) vs. wind that changes over time (∼40 min): within 3 % for windows on different walls, but up to 31 % off for windows on the same wall, so those cases will use the exact method.
Wind speed	Results scale exactly: three times the wind replaces the air three times as fast (within 3 %).
Plausibility	Results match textbook formulas for window ventilation and an ideal, perfectly mixed room.

Where we are

Done: the pipeline runs automatically from a settings file to a table row, and the checks above are complete.
Next: final choice of method, then about 180 simulations (about 2 days of computing on one laptop).
Then: train and test the ML model on the resulting table.